

OPERATING SYSTEMS (CS304)

Unit-wise Syllabus Breakdown

Unit 1 - Introduction & Processes

Topics Covered:

- Introduction to Operating Systems
- Functions and Services of Operating Systems
- Process Concept
- Process States and State Transitions
- Process Control Block (PCB)
- Threads and Multithreading
- System Calls

Unit 2 - CPU Scheduling

Topics Covered:

- CPU Scheduling Concepts
- Scheduling Criteria
- FCFS Scheduling
- SJF Scheduling
- Round Robin Scheduling
- Priority Scheduling
- Synchronization Basics

Unit 3 - Deadlocks

Topics Covered:

- Deadlock Concept
- Deadlock Characterization
- Necessary Conditions for Deadlock
- Resource Allocation Graph
- Deadlock Prevention
- Deadlock Avoidance
- Banker's Algorithm
- Deadlock Detection and Recovery

Unit 4 - Memory Management

Topics Covered:

- Memory Allocation Techniques
- Paging
- Segmentation
- Virtual Memory
- Demand Paging
- Page Replacement Algorithms
- Memory Protection

Unit 5 - Storage & File Systems

Topics Covered:

- File Concepts
- File Allocation Methods

- Directory Structures
- Secondary Storage Management
- Disk Structure
- Disk Scheduling Algorithms
- Storage Management Techniques

Exam Preparation Tips

- Focus on numerical problems in CPU Scheduling.
- Practice Banker's Algorithm thoroughly.
- Understand Paging and Page Replacement algorithms.
- Learn process state diagrams and system calls.
- Revise file allocation methods and disk scheduling techniques.

Summary

This document provides a unit-wise breakdown of the Operating Systems syllabus and highlights the important topics that students should focus on for examinations and interviews.